

Statistical Significance and Runtime Analysis of Hybrid Quantum–Classical Emotion Classification

Pairwise Comparisons of Exact Match Ratio (EMR) and Variational Circuit Complexity

1. Statistical Significance Analysis

1.1 Data and Test Used

All significance tests are computed on the **Exact Match Ratio (EMR)** reported in the main results table of the paper for the EmoNoBa test set of

$$N = 284 \text{ batches} \times 8 \text{ samples/batch} = 2272 \text{ samples.}$$

Each configuration produces a single EMR = $\frac{\text{number of exactly matched samples}}{N}$.

Because the available data is the aggregate EMR per configuration (a single test-set run) rather than per-sample prediction vectors or multiple independent training runs, we compare EMRs as proportions of the shared test set using the **two-proportion z -test** (pooled variance):

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_{\text{pool}}(1 - \hat{p}_{\text{pool}}) \left(\frac{1}{N} + \frac{1}{N} \right)}}, \quad \hat{p}_{\text{pool}} = \frac{x_1 + x_2}{2N},$$

and report the two-sided p -value $p = 2\Phi(-|z|)$. A comparison is deemed *statistically significant* at the $\alpha = 0.05$ level when $p < 0.05$, and the effect is reported with its sign (which configuration is better).

Important caveat. The two-proportion z -test assumes the two predictions are *independent*. Here all configurations share the same test set and are strongly correlated (they often agree on the easy samples), so this test *underestimates* the correlation and may *overstate* significance. For a fully rigorous test one should instead use *McNemar’s test* on per-sample prediction vectors (paired data), or a t -test / Mann–Whitney U test across multiple independent training runs (seeds). The aggregate single-run EMRs available here do not support those tests directly; the z -test below is the most appropriate test possible with the current data and should be interpreted with this limitation in mind.

1.2 IndicBERTv2 Encoder

The classical baseline has EMR = 0.4252.

Classical vs. each quantum configuration

Reading:

- **SEL** is significantly *worse* than the classical baseline at every register size ($p < 0.0001$).

Comparison	Qubits	VQC	EMR	z	p -value	Sig.
Classical	2	SEL	0.4252 vs. 0.0238	32.42	< 0.0001	YES
Classical	4	SEL	0.4252 vs. 0.2148	15.20	< 0.0001	YES
Classical	8	SEL	0.4252 vs. 0.0497	29.74	< 0.0001	YES
Classical	2	Custom 1	0.4252 vs. 0.4230	0.15	0.8807	no
Classical	4	Custom 1	0.4252 vs. 0.4155	0.66	0.5078	no
Classical	8	Custom 1	0.4252 vs. 0.4265	-0.09	0.9294	no
Classical	2	Custom 2	0.4252 vs. 0.1527	20.26	< 0.0001	YES
Classical	4	Custom 2	0.4252 vs. 0.0519	29.52	< 0.0001	YES
Classical	8	Custom 2	0.4252 vs. 0.4577	-2.21	0.0274	YES

Table 1: IndicBERTv2: classical baseline vs. each VQC configuration.

- **Custom 1** is statistically indistinguishable from the classical baseline at 2, 4, and 8 qubits (all $p > 0.05$); the VQC neither helps nor hurts.
- **Custom 2** is significantly *worse* than classical at 2 and 4 qubits ($p < 0.0001$), but significantly *better* at 8 qubits ($p = 0.0274$) — the only quantum configuration under IndicBERTv2 that beats the classical baseline.

Adjacent register-size comparisons within each VQC

VQC	Comparison	EMR	z	p -value	Sig.
SEL	2 qubits vs. 4 qubits	0.0238 vs. 0.2148	-19.86	< 0.0001	YES
SEL	4 qubits vs. 8 qubits	0.2148 vs. 0.0497	16.43	< 0.0001	YES
Custom 1	2 qubits vs. 4 qubits	0.4230 vs. 0.4155	0.51	0.6084	no
Custom 1	4 qubits vs. 8 qubits	0.4155 vs. 0.4265	-0.75	0.4527	no
Custom 2	2 qubits vs. 4 qubits	0.1527 vs. 0.0519	11.21	< 0.0001	YES
Custom 2	4 qubits vs. 8 qubits	0.0519 vs. 0.4577	-31.39	< 0.0001	YES

Table 2: IndicBERTv2: adjacent register-size comparisons within each VQC.

Reading:

- **SEL** changes significantly between every pair of registers (non-monotonic, peaking at 4 qubits).
- **Custom 1** is statistically flat across registers (no significant change with qubit count).
- **Custom 2** changes significantly between every pair, with a dramatic improvement from 4 to 8 qubits ($0.0519 \rightarrow 0.4577$).

1.3 BanglaBERT Encoder

The classical baseline has $\text{EMR} = 0.0000$ (the classical model predicts no positive labels).

Classical vs. each quantum configuration

Reading:

Comparison	Qubits	VQC	EMR	z	p -value	Sig.
Classical	2	SEL	0.0000 vs. 0.0264	-7.85	< 0.0001	YES
Classical	4	SEL	0.0000 vs. 0.3178	-32.53	< 0.0001	YES
Classical	8	SEL	0.0000 vs. 0.0387	-9.56	< 0.0001	YES
Classical	2	Custom 1	0.0000 vs. 0.1505	-20.06	< 0.0001	YES
Classical	4	Custom 1	0.0000 vs. 0.3891	-38.04	< 0.0001	YES
Classical	8	Custom 1	0.0000 vs. 0.3895	-38.07	< 0.0001	YES
Classical	2	Custom 2	0.0000 vs. 0.0000	n/a	n/a [‡]	n/a
Classical	4	Custom 2	0.0000 vs. 0.4071	-39.50	< 0.0001	YES
Classical	8	Custom 2	0.0000 vs. 0.4630	-44.26	< 0.0001	YES

[‡] Both configurations achieve EMR = 0.0000; there is no difference to test.

Table 3: BanglaBERT: classical baseline vs. each VQC configuration.

- Because the classical BanglaBERT baseline collapses to zero EMR, **every** quantum configuration that achieves a non-zero EMR is automatically and significantly better ($p < 0.0001$).
- This reflects the *compensatory* role of the quantum branch discussed in the paper: when the classical embedding is weak, the quantum circuit provides a large and significant gain.
- The 2-qubit Custom 2 also produces zero EMR, so it cannot be distinguished from the classical baseline.

Adjacent register-size comparisons within each VQC

VQC	Comparison	EMR	z	p -value	Sig.
SEL	2 qubits vs. 4 qubits	0.0264 vs. 0.3178	-26.02	< 0.0001	YES
SEL	4 qubits vs. 8 qubits	0.3178 vs. 0.0387	24.58	< 0.0001	YES
Custom 1	2 qubits vs. 4 qubits	0.1505 vs. 0.3891	-18.12	< 0.0001	YES
Custom 1	4 qubits vs. 8 qubits	0.3891 vs. 0.3895	-0.03	0.9779	no
Custom 2	2 qubits vs. 4 qubits	0.0000 vs. 0.4071	-39.50	< 0.0001	YES
Custom 2	4 qubits vs. 8 qubits	0.4071 vs. 0.4630	-3.80	0.0001	YES

Table 4: BanglaBERT: adjacent register-size comparisons within each VQC.

Reading:

- **SEL** peaks significantly at 4 qubits and degrades significantly at 8 qubits.
- **Custom 1** improves significantly from 2 to 4 qubits, then plateaus (4 vs. 8 not significant).
- **Custom 2** improves significantly from 4 to 8 qubits ($p = 0.0001$).

2. Time Complexity and Runtime Analysis

2.1 Asymptotic Complexity of the Quantum Branch

Let n_q denote the number of qubits, L the number of variational blocks (with the associated data re-upload), and B the mini-batch size. The variational quantum circuit is simulated exactly on the

PennyLane `default.qubit` statevector simulator, which represents the state as a complex vector of dimension 2^{n_q} . Applying any single gate to the full statevector costs $\mathcal{O}(2^{n_q})$ time.

For the Custom 2 circuit (one ansatz layer, data re-uploading performed twice), the gate count is

$$\begin{aligned} \text{Encoding block : } & 2 \text{ rotations per qubit} = 2n_q \text{ gates,} \\ \text{Ansatz block : } & n_q \text{ single-qubit Rot} + n_q \text{ CZ} = 2n_q \text{ gates,} \\ \text{Total (2 repetitions) : } & 2(2n_q + 2n_q) = 8n_q \text{ gates.} \end{aligned}$$

Hence a single circuit evaluation has cost

$$T_{\text{circuit}} = \mathcal{O}(n_q 2^{n_q}),$$

and, because `QuantumBottleneck` executes the circuit once per sample in a batch,

$$T_{\text{batch}} = \mathcal{O}(B n_q 2^{n_q}).$$

The dominant cost of a full forward pass is nevertheless the transformer encoder. For a sequence of length S , hidden dimension d , and L_t layers, the self-attention cost is

$$T_{\text{BERT}} = \mathcal{O}(B L_t S^2 d),$$

which, for the large pretrained encoders used here (IndicBERTv2, $\approx 278\text{M}$ parameters), dwarfs the quantum contribution in wall-clock time. The quantum statevector itself is negligible in memory: $16 \cdot 2^{n_q}$ bytes, e.g. 4 KB at $n_q = 8$.

2.2 Measured Runtime

The variational circuit timing was instrumented live during training (“VQC TIMING”), recording the total simulator time, the number of forward circuit calls, the per-batch time ($B = 8$), and the per-sample time. Two eight-qubit **Custom 2** configurations (IndicBERTv2 encoder), two four-qubit **Custom 1** configurations (IndicBERTv2 and BanglaBERT encoders), and two four-qubit **SEL** configurations (IndicBERTv2 and BanglaBERT encoders) are reported.

Quantity	Custom 2, IndicBERTv2 (8q, run A)	Custom 2, IndicBERTv2 (8q, run B)
Total VQC time (s)	13639.8	11729.2
Total VQC time (min)	227.33	195.49
Forward calls	48905	41228
Per batch, $B = 8$ (ms)	278.90	284.50
Per sample (ms)	34.86	35.56

Table 5: Measured variational-circuit runtime for the Custom 2 design (PennyLane statevector simulator, CPU).

Reading:

- The per-sample VQC time for **Custom 2** is ≈ 35 ms and is essentially identical across both eight-qubit runs of the same encoder, confirming that the quantum cost is dominated by the 2^{n_q} statevector dimension and is reproducible run-to-run.
- The ratio of total time to forward calls matches the per-batch figure (e.g. $13639.8 \text{ s} / 48905 = 278.9 \text{ ms}$), so each recorded “forward call” corresponds to one batch of eight circuit executions.

Quantity	Custom 1, IndicBERTv2 (4q)	Custom 1, BanglaBERT (4q)
Total VQC time (s)	5374.5	5238.4
Total VQC time (min)	89.58	87.31
Forward calls	64259	59141
Per batch, $B = 8$ (ms)	83.64	88.58
Per sample (ms)	10.45	11.07

Table 6: Measured variational-circuit runtime for the Custom 1 design at 4 qubits (PennyLane statevector simulator, CPU).

Quantity	SEL, IndicBERTv2 (4q)	SEL, BanglaBERT (4q)
Total VQC time (s)	4029.1	4950.2
Total VQC time (min)	67.15	82.50
Forward calls	59141	59141
Per batch, $B = 8$ (ms)	68.13	83.70
Per sample (ms)	8.52	10.46

Table 7: Measured variational-circuit runtime for the SEL design at 4 qubits (PennyLane statevector simulator, CPU).

- The four-qubit **Custom 1** configurations achieve a per-sample cost of 10.45 ms (IndicBERTv2) and 11.07 ms (BanglaBERT), which is roughly $3\times$ cheaper than the eight-qubit Custom 2 runs (≈ 35 ms) and only marginally slower than the four-qubit SEL configurations (8.52 ms and 10.46 ms respectively). This modest overhead relative to SEL is consistent with Custom 1 employing additional single-qubit rotation gates per layer while operating on the same $2^4 = 16$ -dimensional statevector.
- The quantum simulator is the computational bottleneck for the quantum branch, but in the full hybrid model it is small relative to the transformer forward pass, which is why the total training time is dominated by BERT.
- Ranking by per-sample cost at 4 qubits: SEL (≈ 8.5 – 10.5 ms) $<$ Custom 1 (≈ 10.5 – 11.1 ms) $\ll$ Custom 2 at 8 qubits (≈ 35 ms), consistent with the exponential 2^{n_q} statevector growth dominating when the register doubles from 4 to 8 qubits.

3. Single-Label Analysis

3.1 Dataset Partitioning and Formulation

In multi-label emotion classification, test samples can be partitioned based on label cardinality into single-label instances ($\sum_{i=1}^L y_i = 1$) representing mutually exclusive emotion states and multi-label co-occurrence instances ($\sum_{i=1}^L y_i > 1$). Filtering the EmoNoBa test set for single-label instances yields $N_{\text{single}} = 940$ test samples across six emotion classes: *Happy* ($N = 300, 31.9\%$), *Sad* ($N = 200, 21.3\%$), *Angry* ($N = 200, 21.3\%$), *Disgust* ($N = 100, 10.6\%$), *Surprise* ($N = 80, 8.5\%$), and *Fear* ($N = 60, 6.4\%$). On this subset, strict Exact Match Ratio (EMR), Micro-F1, Precision, and Recall converge to standard multi-class classification accuracy.

3.2 Single-Label Exact Match Ratio (EMR) Benchmark

Table 8 summarizes the single-label Exact Match Ratio (EMR) scores across the three investigated VQC notebook implementations: the 4-qubit SEL reference design (`4qb_sel.ipynb`), the 4-qubit Custom 1 cross-coupling design (`Biswanath_4_qubit.ipynb`), and the 8-qubit Custom 2 data re-uploading design (`tathagata_main_codebase_v2_8_qubit.ipynb`).

Encoder	VQC Architecture	Qubits	Exact Match Ratio (EMR)
IndicBERTv2	SEL	4	0.4713
IndicBERTv2	Custom 1	4	0.4521
IndicBERTv2	Custom 2	8	0.5479
BanglaBERT	SEL	4	0.4500
BanglaBERT	Custom 1	4	0.3191
BanglaBERT	Custom 2	8	0.5330

Table 8: Single-label Exact Match Ratio (EMR) across evaluated VQC notebook architectures ($N_{\text{single}} = 940$).

3.3 Architectural Investigation and Comparative Analysis

Analysis of the three VQC notebook implementations reveals how encoding complexity, entanglement topology, and measurement strategies dictate performance on single-label decision boundaries:

- **SEL Architecture (`4qb_sel.ipynb`):** Combines a single-pass $R_X(x_i)$ angle embedding with a single layer of PennyLane’s `StronglyEntanglingLayers` (applying general $Rot(\phi, \theta, \omega)$ rotations and default CNOT entanglement) and 4-channel Pauli- Z expectation measurement ($\langle Z_i \rangle$). On single-label instances, SEL achieves moderate EMR (0.4713 on IndicBERTv2, 0.4500 on BanglaBERT) but exhibits extreme majority-class bias, predicting non-zero recall only for high-support categories (*Happy* and *Angry*) while completely failing on minority classes (*Disgust*, *Surprise*, and *Fear* all record 0.0000 recall).
- **Custom 1 Architecture (`Biswanath_4_qubit.ipynb`):** Implements an $R_Y(x_i)$ angle embedding paired with a two-layer ansatz featuring distinct CNOT entangling topologies: ring coupling ($0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 0$) in Layer 1 and cross-coupling ($0 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 1, 3 \rightarrow 0$) in Layer 2. Despite the deeper CNOT entanglement, the absence of data re-uploading limits feature separability. Under BanglaBERT, Custom 1 experiences complete majority-class collapse, predicting the dominant *Happy* label for all samples (recall = 1.0000 on *Happy*, 0.0000 on all other classes), resulting in an EMR of 0.3191 (matching the test set proportion 300/940).
- **Custom 2 Architecture (`tathagata_main_codebase_v2_8_qubit.ipynb`):** Features a data re-uploading scheme ($U(x) \rightarrow V(\theta) \rightarrow U(x) \rightarrow V(\theta)$) operating on an 8-qubit register. Input features are transformed using learnable per-qubit phase (ϕ_q) and scale (λ_q) parameters into paired R_X and R_Y rotations:

$$x_{RX,q} = \lambda_q(\cos \phi_q \cdot x_{2q} - \sin \phi_q \cdot x_{2q+1}), \quad x_{RY,q} = \lambda_q(\sin \phi_q \cdot x_{2q} + \cos \phi_q \cdot x_{2q+1}).$$

The ansatz combines general *Rot* transformations with ring Controlled- Z (CZ) entangling gates ($q \rightarrow (q + 1) \bmod 8$) and an expanded 16-channel Pauli- Z and Pauli- X readout ($\langle Z_q \rangle, \langle X_q \rangle$). Data re-uploading and phase-scale mixing allow Custom 2 to construct highly non-linear decision

boundaries, achieving peak single-label EMR performance (0.5479 for IndicBERTv2, 0.5330 for BanglaBERT) and preventing class-prediction collapse across both language encoders.